

Chapter 4 Basic equations of steady total flow

Section 1 Method of total flow analysis

Section 2 Continuity equation

Section 3 Bernoulli equation for steady total flow

Section 4 Momentum equation for steady total flow

Chapter summary

Chapter 4 Basic equations of steady total flow

Key study points:

- Grasp the total-flow analysis approach for fluid motion
- Understand the continuity equation, Bernoulli equation and momentum equation for steady total flow, as well as their applications



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Section 1 Method of total flow analysis

- 👉 1. Basic concept
- 2. Method of total flow analysis

Section 1: Method of total flow analysis

1. Concept

Element Flow

- Stream filament with an infinitely small cross section
- The limit of element flow is streamline
- The motion of each point on the section is **the same**

Total Flow

- Stream filament with a finite-size cross-section
- Consisting of numerous element flows
- The motion of each point on the section is generally **different**

Control cross section

- Flow cross-section with in/out flows on the surface of a control volume
- Control volume: volume occupied by the water between Sections 1-1 and 2-2.

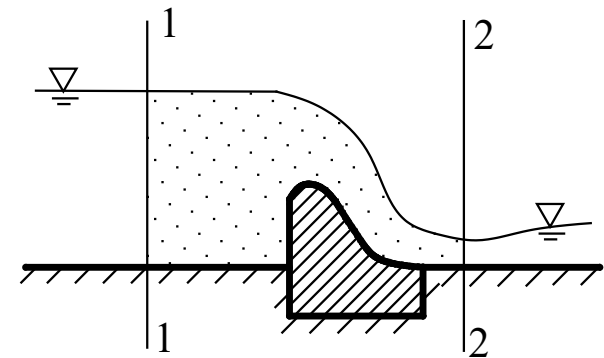


Fig 4.1 Control volume control cross section of **total flow**

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Section 1 Method of total flow analysis

1. Basic concept
-  2. Method of total flow analysis

Chapter 4 Basic equations of steady total flow

2. Method of total flow analysis

- (1) Starting from element flow
- (2) Selecting control sections in gradually-varied flow areas
- (3) Taking the cross-sectional average of certain physical quantities

Chapter 4 Basic equation of steady total flow

(1) Starting from element flow

- The total flow is the sum of numerous element flows
- Analyze the element flow before total flow
- Integrate the element flow formulations to get those for total flow



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(2) Selecting control sections in gradually-varied flow areas

- In total flow analysis, the control section is always selected in areas with **gradually-varied flow**, or its limit case, i.e. **uniform flow**

Think: Why control sections need to be in areas with gradually-varied flow?



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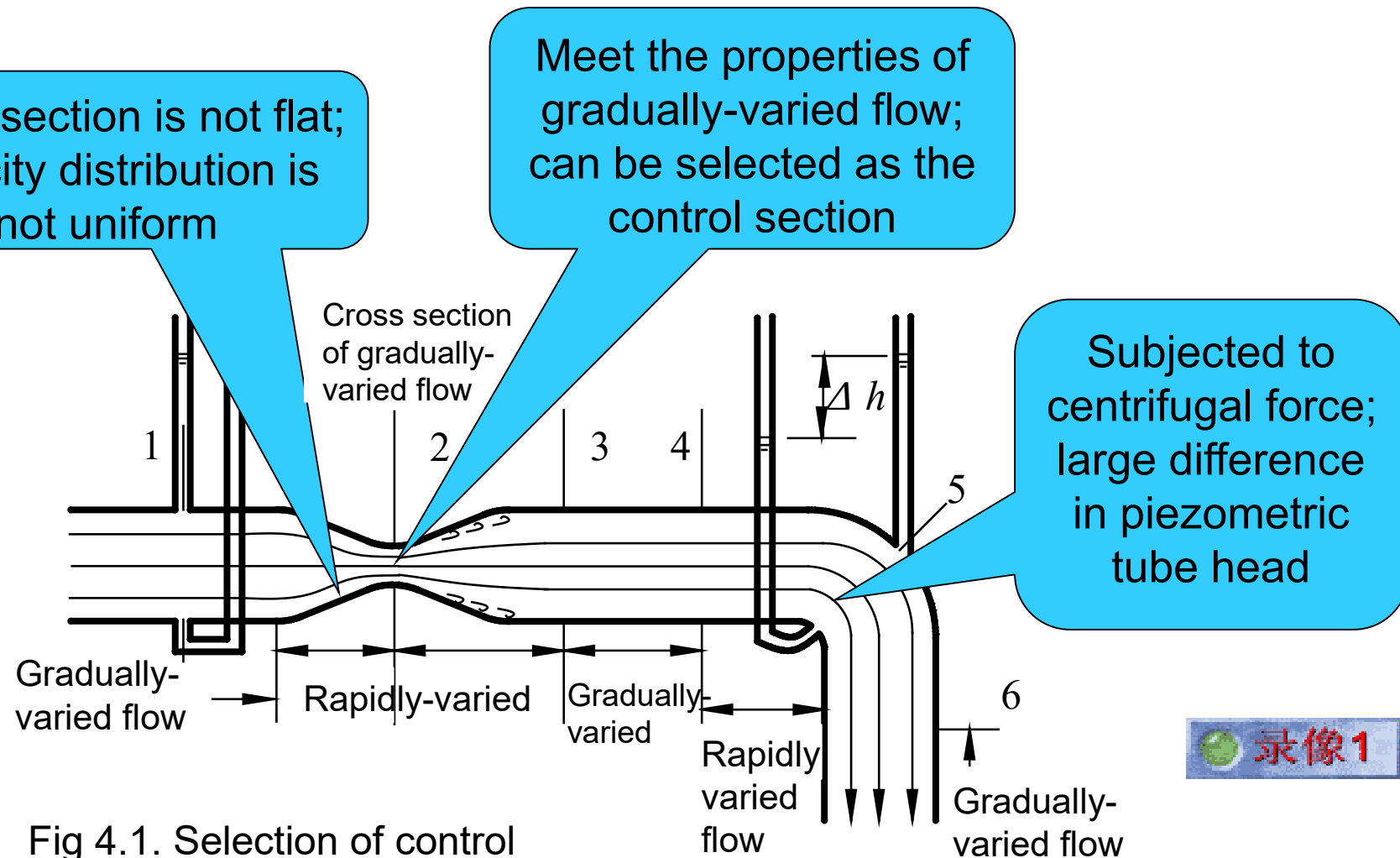
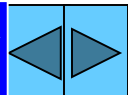


Fig 4.1. Selection of control sections of total flow



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(3) Taking the cross-sectional average of physical quantities

- 👉 1) Cross-section mean velocity
- 2) Momentum and momentum correction factor
- 3) Kinetic energy and kinetic energy correction factor

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1) Cross-Section Mean Velocity

The velocity at a point of fluid flow is called **the point velocity**, usually denoted by u .

Mean velocity v
$$v = \frac{\int_A u dA}{A} = \frac{q_V}{A}$$

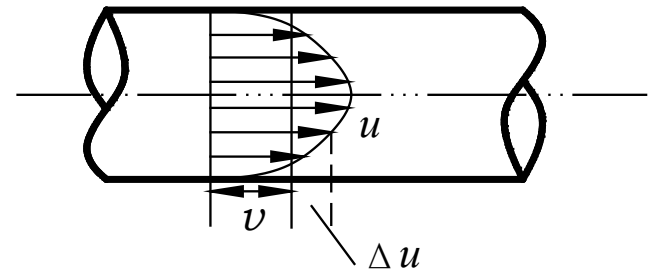


Fig 4.3 Mean velocity

Geometric meaning: The volume of a cylinder with bottom A and height v equal to the volume enclosed by the velocity distribution curve and the flow cross-section

$$\int_A u dA$$



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(3) Taking the cross-sectional average of physical quantities

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2) Momentum and momentum correction factor

Momentum: a measure of the movement of an object; an important physical quantity describing the mechanical movements of objects.

The momentum of element flow through the cross section within time Δt :

$$d\vec{K} = dm\vec{u} = (\rho u dA \cdot \Delta t) \vec{u} = \rho u \vec{u} dA \cdot \Delta t$$

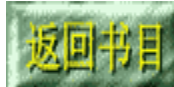
The momentum of the total flow through the whole cross section is:

$$\sum d\vec{K} = \int_A \rho u \vec{u} dA \cdot \Delta t = \left[\int_A \rho u dA u \right] \vec{i} \cdot \Delta t = \beta \rho q_v v \cdot \vec{i} \cdot \Delta t = \beta \rho q_v \vec{v} \cdot \Delta t$$

To express the integral in the above equation by the average cross-sectional velocity v , a **momentum correction factor** β is introduced:

$$\beta = \frac{\int_A u^2 dA}{q_v v} = \frac{\int_A u^2 dA}{A v^2}$$

The value of β depends on the velocity distribution on the total-flow cross section. For generally uniform velocity distribution, $\beta = 1.02 \sim 1.05$. $\beta = 1.0$ is often adopted in calculation.



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(3) Taking the cross-sectional average of physical quantities

- 1) Cross-section mean velocity
- 2) Momentum and momentum correction factor
- 👉 3) Kinetic energy and kinetic energy correction factor

Chapter 4 Basic equation of steady total flow

3) Kinetic energy and kinetic energy correction factor

Kinetic energy: The energy of an object as a result of mechanical motion

The kinetic energy of mass $\rho u dA$ flowing through an infinitesimal area dA , per unit time, is

$$dE_k = \frac{1}{2} dm \cdot u^2 = \frac{1}{2} \rho u^3 dA$$

The kinetic energy of the total flow of incompressible fluid flowing through the cross section, per unit time

$$\sum dE_k = \frac{1}{2} \rho \int_A u^3 dA = \rho q_v \frac{\alpha v^2}{2}$$

To express the integral in the above equation by the averaged cross-sectional velocity v , a **kinetic energy correction factor** α is introduced

$$\alpha = \frac{\int_A u^3 dA}{q_v v^2} = \frac{\int_A u^3 dA}{A v^3}$$

- The value of α depends on the velocity distribution on the cross section
- For generally uniform velocity distribution, $\alpha = 1.05 \sim 1.1$
- α is usually taken as 1.0 in engineering calculation

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Section 2 Continuity equation

- ➡ 1. Continuity equation for steady incompressible total flow
- 2. Total-flow continuity equation for bifurcated flow

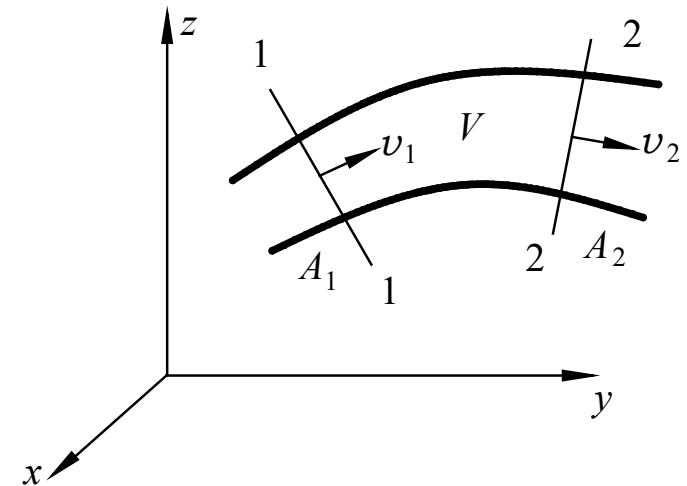
1. Continuity equation for steady incompressible total flow

$$\int_V \left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} \right) dV = 0$$

By Gauss Theorem:
$$\int_V \left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} \right) dV = \int_A u_n dA$$

$$\therefore \int_A u_n dA = 0 \implies \int_{A_1} u_1 dA = \int_{A_2} u_2 dA$$

$$\implies v_1 A_1 = v_2 A_2 \quad \text{or} \quad q_{V1} = q_{V2}$$



Physical meaning : For **incompressible fluid**, the cross-sectional average velocity is **inversely proportional** to the cross-sectional area. That is, the flow velocity is large where the streamlines are dense, and vice versa.

Scope of application : **incompressible fluid** within fixed boundaries, including **steady flow, unsteady flow, ideal flow, real flow.**



Section 2 Continuity equation

1. Continuity equation for steady incompressible total flow

 2. Total-flow continuity equation for bifurcated flow

2. Total-flow continuity equation of bifurcated flow

$$v_1 A_1 = v_2 A_2 + v_3 A_3$$

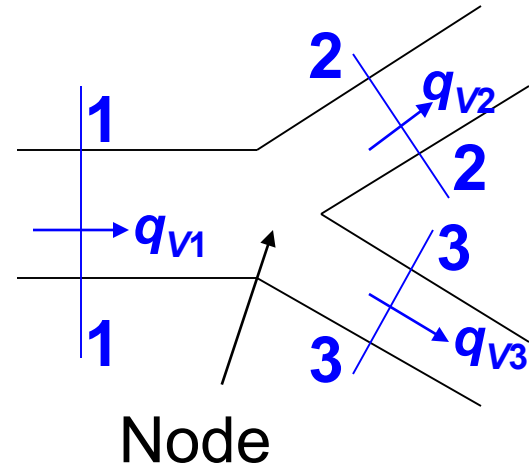
or $q_{V1} = q_{V2} + q_{V3}$

Node continuity equation:

$$\sum_{i=1}^n q_{Vi} = 0$$

n ——Number of branch pipe。

Flow out of the node is “+”; flow into the node is “—”。



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